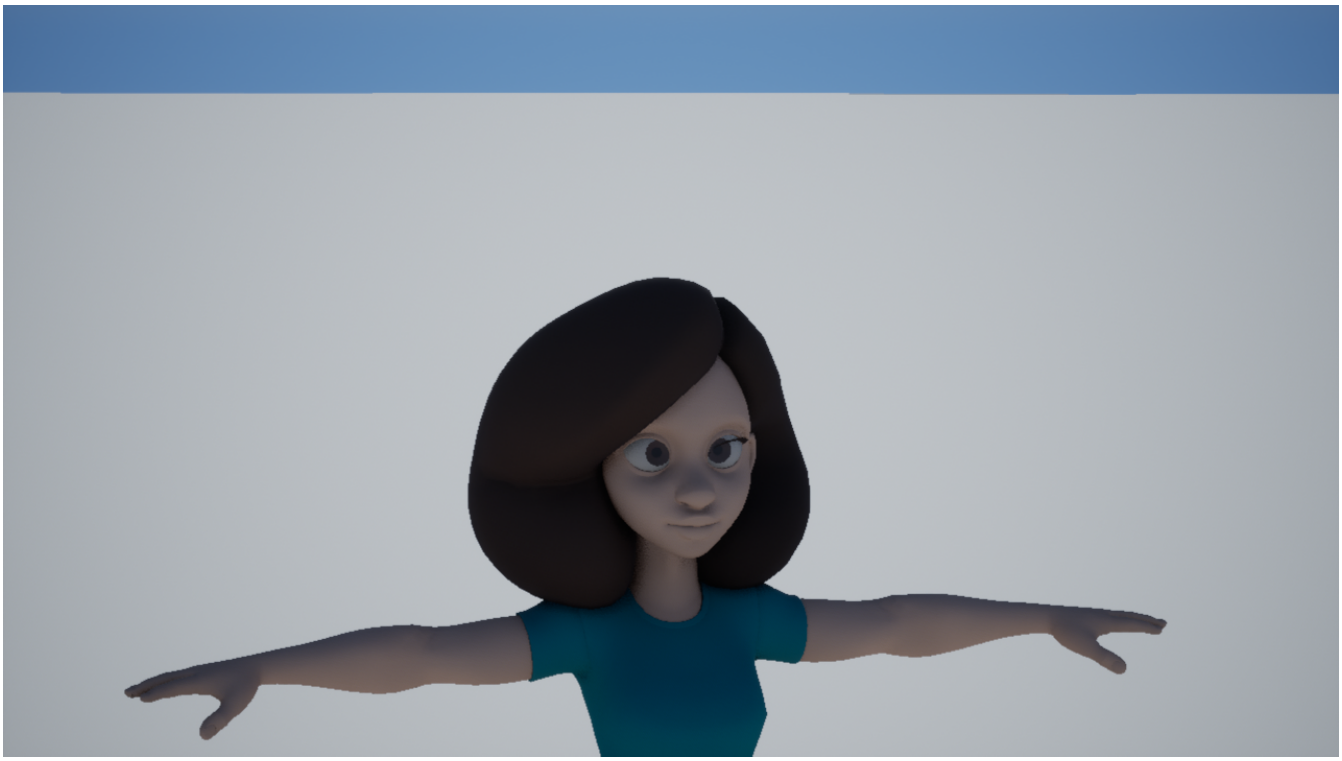
HUMBER FILM 304 / CLASS REVIEW EDITION

Toon Materials & Character Handoff

A production bible for the Humber Toon Spine project



Dana material and lighting preview. Static source pose; this image does not prove rig deformation.

Prepared for the Film 304 group review | 16 October 2026

01 / START HERE

Project snapshot

This is a working guide to the Unreal toon materials, the film look, and the character handoff. It records what is built, what has been seen in a render, and what still needs a clean export.

The build rule
Material assets are generated from the Python builders in this repository. Change the builder, rebuild the asset, then read it back in Unreal. An editor-only graph edit will be lost on the next rebuild.

Step	What the artist chooses	Where it lives
1. Pick a family	Skin, hair, face, glass, cloth, ground, or sky.	M_* master
2. Make a look	Tint, roughness, line weight, or a surface pattern.	MI_* instance
3. Set the bands	Choose the light and shadow response for the subject.	TP_* profile
4. Review the shot	Check the look under the shot's actual light and grade.	Saved still + review notes

Working rules

- One person owns each Unreal binary while it is being edited. Material instances are the place for look variations.
- Keep changes in the builder that creates the asset. Record a still and the build report with the change.
- Use the SM6 workstation for final shader judgement. The Oct 8 laptop record says Substrate surfaces rendered black on that machine.

Status in one sentence
The six film utilities and static deep-night preset have build and fresh-load compile records; visual approval of the sky and a final Dana deformation export remain open.

02 / LOOK DEVELOPMENT

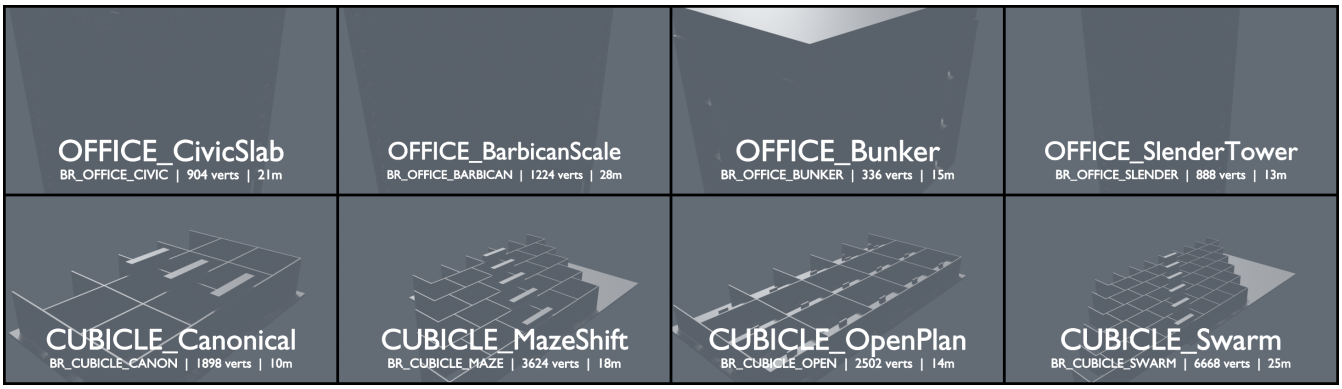
The material map

The spine keeps light response in a master, subject bands in a profile, and the visible variation in an instance. Outlines and the final film grade have their own place in the stack.

Surface	Start with	Use it for
Shared surfaces	M_Master_Toon_Universal + MI_Toon_*	Office props, masonry, and surfaces that share the main band response.
Character body	M_Master_Toon_Character	Clothing and broad body shading.
Face and hair	M_Master_Toon_Face / M_Master_Toon_Hair	Face shadow shape, scalp, and hair response.
Eyes and cornea	DanaFilm eye instances	Separate sclera, iris, eye interior, and cornea controls.
Landscape	M_Master_Toon_Landscape	Ground and broad terrain variation.
Sky	M_Master_Toon_Sky + MI_Toon_Sky_DeepNight	One fixed deep-night look.
Final grade	M_Master_Toon_PostComposite	One shared neutral grade, before per-shot tint.
Line work	M_Outline_InvertedHull + MI_Outline_Thin/Heavy	Thin lines on characters; heavier lines on large architecture.

Keep the image readable

- Keep the darkest toon band above pure black so shadowed shapes still read.
- Use the toon profile to shape the light bands; use the material instance for color and surface detail.
- Keep one final grade in the post-composite material. Avoid stacking separate grades shot by shot.



Office and cubicle model families from the class staging scaffold.

03 / REUSABLE CONTROLS

Six film tools and one sky preset

Each utility is quiet at its default. Open a control only when the shot needs it, then judge the change in the final lighting.

Function	Useful for	Artist controls and safe start
MF_NormalAdjust	Soften or sharpen toon bands across curved forms.	NormalStrength 1.0 is identity; below 1 softens, above 1 tightens.
MF_SurfaceWear	Small worn patches and crack marks on walls or floors.	WearStrength 0; CrackStrength 0. WearScale 0.02 is about a 50 cm tile. CrackWidth 0.01-0.08.
MF_DF_ContactBlend	Dirt and roughness close to contact points.	DFContactStrength 0. Tint and roughness stay unchanged until strength is raised.
MF_Impasto	Raised, directional paint strokes.	ImpastoStrength 0. BrushScale 0.02-0.09; 0.045 gives broad strokes.
MF_ClothWindDrape	Subtle motion in cloth and stage fabric.	WindStrength 0-1 cm; 0.05-0.15 for a drift, 0.2-0.35 for a walk breeze. Speed 0.2-1.5 cycles/sec.
MF_FilmGrade	A small neutral finish across the image.	Contrast 1, Warmth 0, Saturation 1, Lift 0. GradeTint remains the shot tint.

Deep-night sky
MI_Toon_Sky_DeepNight sits on M_Master_Toon_Sky. It uses an indigo-black zenith, a city-glow horizon, four value bands, and sparse stars. It is a fixed preset; there is no time-of-day control.

Keep iridescence, sparkle, vein glow, and radial rings out of this film set. Add a new function only when a shot has a named, visible need for it.

04 / FILES AND HANDOFF

Names that travel

Inside Unreal, use the prefixes in Docs/CONTENT_CONVENTIONS.md. In the Humber staging folders, use the role prefixes from GROUP_STAGING_GUIDE.md. The staging name describes the incoming file; the Unreal name describes the asset people will use.

Unreal asset	Prefix	Example
Master material	M_	M_Master_Toon_Universal
Material instance	MI_	MI_Toon_Stone
Material function	MF_	MF_RampLUT
Toon profile	TP_	TP_Hero
Texture	T_	T_Concrete_N
Static mesh	SM_	SM_Brutalist_Lot_A
Blueprint	BP_	BP_Brutalist_Kit
Level / sequence	L_ / LS_	L_Toon_Lookdev / LS_Shot010

Staging folder examples

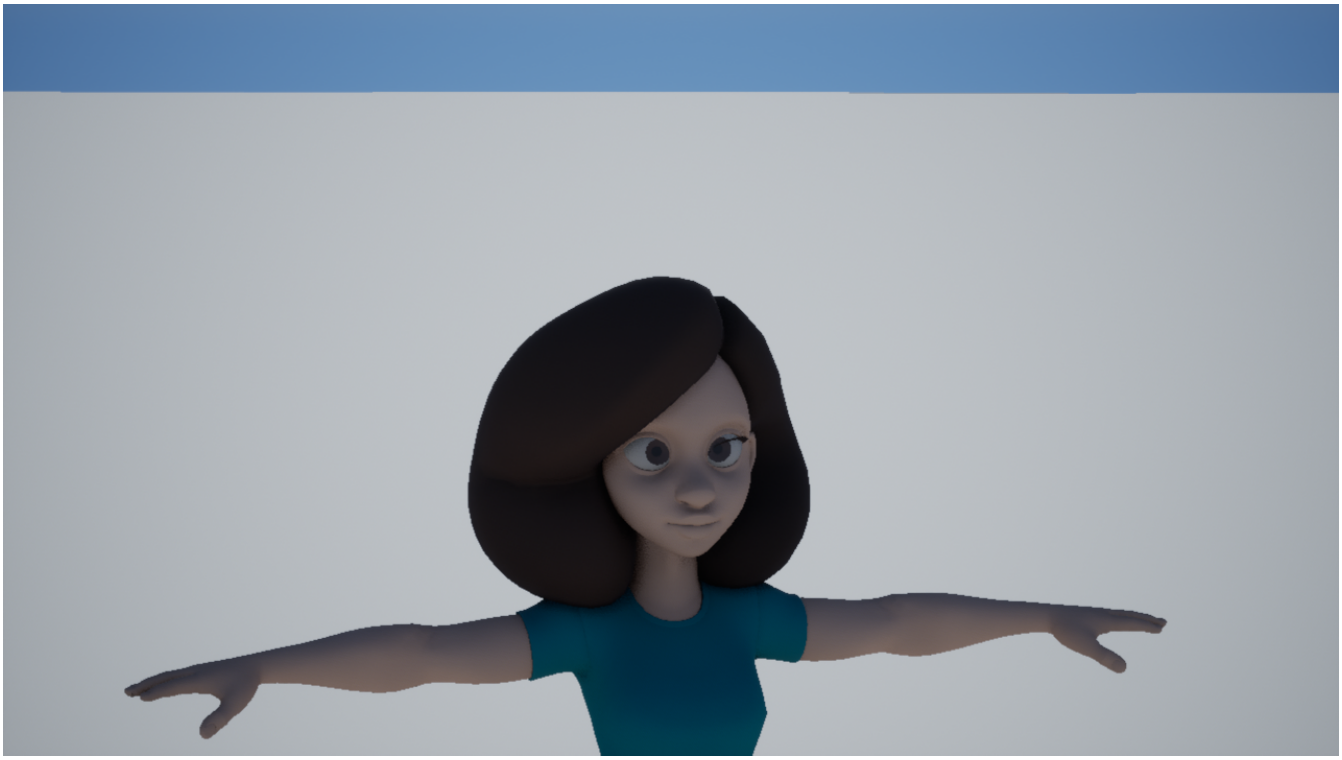
Incoming work	Folder and prefix
Character model	02_Assets/Models/Characters/CH_*
Character rig	02_Assets/Rigs/Character/RIG_CH_*
Environment model	02_Assets/Models/Environments/EN_*
Prop model	02_Assets/Models/Props/PR_*
Material work	02_Assets/Materials/M_*, MI_*, TP_*, T_*

- For imported textures, use _D for diffuse, _N for normal, _R for roughness, and _M for masks.
- Use a word for a real variant, such as _Thin or _Heavy. Do not make Stone2 or final_final.
- The staging guide uses _v01 for versioned source drops. Keep Unreal asset names stable and meaningful.
- The local UE naming guide does not set a SkeletalMesh prefix. Agree that prefix with the group before the final Dana import.

Before importing
Confirm the file owner, source path, units, up axis, material slots, skin weights, and intended Unreal name. Keep the source FBX beside its import notes.

05 / CHARACTER HANDOFF

Dana: what the preview proves



Current daylight preview. It is a static pose and the upper lash line is not reading in this image.

Readback	Current finding
Material stage	The DanaFilm preview has one Dana actor and 11 assigned material slots in the build record.
Body deformation	The source FBX body has zero skin clusters. The still does not prove a working body or face rig.
Upper/lower lashes	39 vertices and 48 triangles per side; all vertices weighted to FaceUpperRegion_M.
Visibility	Both intended lash transforms and their FaceDeformationSystem parent are exported hidden.
Alternate squint pair	The two squint meshes have no skin weights. Keep them out of the render export.
Material slot	The lash geometry uses danaHair_mat. The slot is mapped to the hair instance; the missing lash line is not a shader assignment fault.

Next source export
In Maya, unhide the intended upper/lower pair, repair the face/head export hierarchy so its influence bones belong to the same exported skeleton as the body, and add the body skinCluster. Then test blink, close, look-up, and squint before replacing the Unreal preview.

A filtered FBX preview preserved the lashes, but the body still had no skin cluster. That file is useful for diagnosis, not as the production rig export.

06 / SHOT-FACING NOTES

Materials by part of the film

These are material references from the Office Spider shot plan. Use the storyboard and the current shot owner to confirm any change before promoting a look.

Shot range	Material work	Status in the handoff notes
SH010, SH030	Brutalist exterior, landscape, stone, foliage, sky, and outline weight.	Exterior kit is staged. Deep-night sky still needs its SH010/SH020 review.
SH020-SH060	Cubicle plastic, carpet, laminate, paper, steel, screen, face, hair, and glass.	Interior pieces are staged. Character and spider rig binds remain a dependency.
SH070-SH120	Kitchen and hallway surfaces, glass, coffee props, spider and character materials.	Work is deferred until the required rigs and set build-out are ready.

Examples from the prop shelf

- Paper uses a matte finish and paper grain. Donut boxes use the cardboard surface.
- The coffee machine uses brushed metal; VCT floor grout and carpet loops should still read at the final shot scale.
- Frosted glass belongs on the door glazing. Keep outlines thinner on characters and heavier on architecture.

A material check is visual
A green build report says the graph loaded and the parameters exist. It does not tell us whether the band, outline, shadow, or texture reads well in the shot.

07 / GATES AND NEXT STEPS

Review before calling it done

Gate	Evidence on record	Still needed
Material build	Six film utilities and the static sky preset were built through Python; the Oct 7 fresh-process compile record was clean.	Refresh the gate after any builder changes.
Look review	Dana day and night stills exist; the capture report records three fresh images.	Judge final Substrate materials on the SM6 workstation. One Oct 13 render-queue attempt reported a stale image.
Dana rig	FBX anatomy, lash filter, and pose-clean reports are saved.	Maya source export with body skin weights and a working head/face chain.
Sky preset	Static parameters and preset are present in the spine.	View it in SH010 and SH020, then approve or adjust it.

Next four actions

- Re-export Dana from Maya with one body skinCluster, visible upper/lower lashes, and the final face/head deformation chain.
- Import the export into a clean preview, check one character only, and read back every material slot.
- Refresh the stills on the SM6 workstation and review the sky preset in SH010 and SH020.
- After those reviews, stage the next shot range when the kitchen, hallway, and remaining rigs are ready.

Records used

Docs/CONTENT_CONVENTIONS.md; Docs/GROUP_WORKFLOW.md; Humber_FinalYear_Prep/GROUP_STAGING_GUIDE.md; Docs/TOON_SPINE.md; Docs/MELODIA_TOON_CONVERGENCE.md; Docs/HANDOFF_2026-10-08_NEXT_PHASE.md; Docs/OFFICE_SPIDER_SHOT_PLAN.md; Showcase/Dana/README.md; Saved/Audit/dana_fbx_raw_anatomy.json; Saved/Audit/dana_fbx_geometry_filter_Dana_Film_EyelashFiltered.json; Saved/Audit/dana_fbx_deform_prune_Dana_Film_EyelashPreview.json; Saved/Audit/dana_refined_uds_build.json; Saved/Audit/dana_rig_preview_capture.json; Saved/Audit/dana_refined_uds_render_results.json.